

MAFIKENG, South Africa

WMO: 682420

Lat: 25.8036S

Lon: 25.5428E

Elev: 4203

StdP: 12.60

Time Zone: 2.00 (E02)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	34.0	37.1	3.6	7.7	66.8	8.7	10.0	62.0	24.4	61.3	21.4	59.4	6.1	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	22.7	94.9	61.2	92.5	61.4	90.0	61.3	69.2	80.8	68.2	79.9	67.3	79.2	9.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	113.5	72.3	64.8	107.7	71.7	64.2	105.2	71.5	36.1	79.8	35.2	80.3	34.4	79.2	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
23.2	20.2	17.8	DB	29.0	98.2	2.6	3.1	27.2	100.4	25.7	102.2	24.2	103.9	22.4	106.1	
			WB	25.4	71.5	2.4	2.1	23.7	73.0	22.3	74.2	21.0	75.4	19.3	76.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.8	75.0	74.3	71.4	65.7	60.0	55.2	54.5	59.5	66.6	71.4	73.3	75.0
	DBStd	8.95	4.41	3.89	4.12	4.76	4.82	4.50	4.59	5.93	6.11	5.93	5.61	4.77
	HDD50	36	0	0	0	0	4	11	15	6	0	0	0	0
	HDD65	1095	1	0	4	46	161	294	327	184	54	16	7	1
	CDD50	6159	774	680	663	470	316	167	153	302	498	663	698	775
	CDD65	1742	310	260	203	66	8	0	0	14	102	214	254	311
	CDH74	18289	2826	2283	1774	731	270	55	44	372	1439	2565	2807	3123
	CDH80	7130	1191	892	599	154	28	5	1	52	500	1110	1229	1369
(14) Wind	WSAvg	8.1	8.3	7.5	7.0	6.8	6.4	7.0	7.3	8.8	9.4	10.0	9.9	9.1
(15) Precipitation	PrecAvg	22.1	4.6	3.3	3.0	2.0	0.6	0.2	0.1	0.2	0.6	1.7	2.7	3.5
	PrecMax	32.6	12.5	9.5	7.4	7.4	3.1	1.7	0.5	2.2	2.7	5.3	8.3	9.3
	PrecMin	13.8	1.1	0.6	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.1
	PrecStd	4.6	2.4	1.9	1.8	1.7	0.8	0.4	0.2	0.4	0.7	1.3	2.0	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	97.2	95.0	93.2	87.8	83.8	78.9	78.2	84.5	91.6	95.4	96.5	98.1
		MCWB	63.0	62.5	61.7	60.4	55.6	52.5	51.1	55.0	57.1	58.9	60.0	62.6
	2%	DB	93.0	91.5	89.4	84.2	79.2	75.1	75.0	80.9	88.6	92.9	93.0	93.5
		MCWB	63.7	63.2	62.5	59.7	55.1	51.2	50.1	53.4	56.4	58.4	60.1	63.0
	5%	DB	89.7	88.5	86.3	81.0	77.1	72.2	71.9	78.5	86.1	89.8	89.9	90.2
		MCWB	64.0	63.6	63.0	60.0	54.4	50.2	49.6	52.7	55.7	58.4	60.3	63.0
	10%	DB	86.4	85.9	83.3	78.6	74.7	69.8	69.5	75.2	82.7	86.5	87.4	87.5
		MCWB	64.3	64.5	62.9	59.6	53.7	49.7	49.0	51.6	55.0	57.8	60.7	63.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.7	70.6	69.2	66.5	61.0	56.3	54.5	58.4	62.1	65.7	67.8	69.3
		MCDB	82.1	80.2	79.9	78.1	74.0	69.8	70.2	76.6	80.5	81.8	81.7	82.6
	2%	WB	69.2	69.2	67.6	64.5	58.6	54.0	52.7	56.4	59.9	63.6	66.3	68.1
		MCDB	81.0	79.6	78.4	76.3	71.9	68.6	68.1	74.9	79.7	81.0	80.9	82.0
	5%	WB	68.1	68.1	66.6	63.2	57.0	52.6	51.2	54.7	58.2	62.1	64.9	67.1
		MCDB	79.3	78.8	77.8	74.8	70.5	67.5	67.6	72.6	78.7	79.9	80.3	81.0
	10%	WB	67.1	67.0	65.6	61.7	55.6	51.2	50.1	53.1	56.7	60.9	63.8	66.2
		MCDB	78.5	78.2	76.8	73.1	69.4	66.1	67.1	70.9	77.2	78.8	79.1	80.0
(35) Mean Daily Temperature Range	5% DB	MDBR	21.3	21.2	21.9	23.3	27.0	28.5	28.4	29.1	28.6	26.2	24.6	22.7
		MCDBR	26.4	26.2	27.0	27.0	30.6	31.7	31.5	31.5	31.7	30.0	28.4	27.1
	5% WB	MCWBR	8.4	8.8	9.3	10.2	12.9	14.4	14.7	14.1	13.5	11.8	10.0	9.1
		MCWBR	19.9	20.0	20.7	21.7	24.6	26.4	27.3	27.6	26.8	24.3	23.1	22.1
(40) Clear-Sky Solar Irradiance	taub		0.365	0.372	0.356	0.323	0.285	0.269	0.268	0.322	0.369	0.394	0.346	0.353
		taud	2.398	2.376	2.406	2.456	2.534	2.549	2.526	2.365	2.238	2.213	2.403	2.423
	Ebn at Noon		310	303	298	294	294	293	298	290	289	293	314	315
		Edh at Noon	41	41	37	32	27	25	27	35	43	47	40	40
(44) All-Sky Solar Radiation	RadAvg		2238	2058	1853	1563	1414	1273	1387	1644	1989	2167	2298	2321
(45)	RadStd		226	185	148	132	57	53	32	74	75	118	178	143

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page